

Quadratic functions

Shape → read the turning point → complete the square → sign → model

01 Start with the shape

Look at a , the coefficient of x^2 in $y = ax^2 + bx + c$.

$$a > 0$$

SMILE = minimum

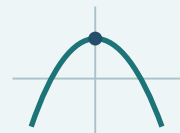
Lowest turning point



$$a < 0$$

FROWN = maximum

Highest turning point



02 Read the turning point

$$y = a(x - h)^2 + k$$

turning point = (h, k)

INSIDE $(x - h)^2$ → x-coordinate

Set bracket = 0; reverse the number's sign.

OUTSIDE + k → y-coordinate

Keep k as it is.

Example: $y = -(x + 3)^2 + 5$ → turning point $(-3, 5)$, maximum 5.

03 Complete the square when needed

Factor out the coefficient of x^2 from the x terms first.

$$f(x) = 2x^2 - 8x + 5$$

$$= 2(x^2 - 4x) + 5$$

$$= 2[(x - 2)^2 - 4] + 5$$

$$= 2(x - 2)^2 - 3$$

READ OFF

TP = $(2, -3)$

minimum = -3

range: $y \geq -3$

The square term is never negative; with $a > 0$, this is a minimum.

04 Always positive or always negative?

Look at the minimum or maximum value k , not just the sign of a .

ALWAYS POSITIVE

$a > 0$ and $k > 0$

ALWAYS NEGATIVE

$a < 0$ and $k < 0$

If $k = 0$, the graph touches $y = 0$; it is not strictly positive or negative.

05 Use it as a model

A rectangle has sides x cm and $(12 - x)$ cm. What is its greatest area?

$$A(x) = x(12 - x) = -(x - 6)^2 + 36$$

$0 < x < 12$; maximum area = 36 cm^2 when $x = 6$ cm. Check domain and units.